

Techno-Economic Evaluation and Investment Decision Analysis of Industrial ORC Systems: A Business Analytics Perspective

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PhD Application Research Portfolio

Abstract. Industrial facilities often release unused thermal energy during production. Organic Rankine Cycle (ORC) systems can convert part of this waste heat into electricity, but adoption depends on whether the expected financial benefits justify the capital cost. This paper develops a business analytics framework for evaluating industrial ORC investments. The study uses annual revenue estimation, Return on Investment (ROI), Net Present Value (NPV), payback period, scenario analysis, and sensitivity analysis. The research intentionally focuses on investment decision support rather than thermodynamic equipment design. Using a simplified 200 kW industrial ORC case, the analysis shows that electricity price, annual operating hours, and capital expenditure are the most important drivers of financial performance. The results indicate that ORC projects can become attractive when electricity prices are sufficiently high and equipment utilization is stable. The paper demonstrates how quantitative business analytics methods can help managers evaluate long-term industrial energy investments under uncertainty.

Keywords: Business analytics; ORC; waste heat recovery; investment analysis; NPV; ROI; sensitivity analysis

1. Introduction

Industrial companies continuously search for methods to reduce energy expenses, improve operating efficiency, and support sustainability goals. Waste heat recovery is one practical opportunity because many manufacturing processes release heat that is not directly used after production. Organic Rankine Cycle systems provide a way to convert low- and medium-temperature waste heat into electricity. However, for managers and investors, the central question is not only whether the technology can work, but whether it creates sufficient economic value.

This study frames ORC adoption as a capital investment decision. The project does not attempt to optimize turbine design, working fluid selection, or heat exchanger architecture. Instead, it evaluates whether projected electricity savings can justify the initial cost of installing an ORC system. This positioning is important for business analytics and data science research because many real-world technology decisions require translating technical potential into measurable financial outcomes.

The research question is: Under what economic conditions can an industrial ORC system generate attractive investment returns? To answer this question, the paper applies a structured quantitative framework based on revenue estimation, ROI, NPV, payback period, scenario analysis, and sensitivity analysis.

2. Background and Related Work

Prior ORC studies have shown that ORC technology is commonly considered a promising approach for converting low- and medium-temperature thermal energy into electricity. Review work by Velez et al. describes ORC technology from technical, economic, and market perspectives, while Quoilin et al. present experimental and modeling work on an ORC prototype using a scroll expander. Lecompte et al. further review ORC architectures for waste heat recovery and discuss the need for cost-effective

design choices.

Although engineering studies are essential, they do not fully answer the investment question faced by firms. A technically feasible ORC system may still be unattractive if the electricity price is too low, annual operating hours are limited, or the capital cost is excessive. Therefore, financial evaluation methods are needed to connect engineering potential to managerial decision making.

Business analytics contributes to this problem by providing a structured way to evaluate investment risk. Instead of relying on a single optimistic assumption, scenario analysis compares conservative, base, and optimistic cases. Sensitivity analysis then identifies which assumptions drive the largest changes in NPV. This makes the project more useful for management decisions and easier to communicate to non-engineering stakeholders.

3. Research Framework

The analysis follows a simple four-step framework. First, the annual electricity generation of the ORC system is estimated from installed capacity and operating hours. Second, annual revenue is calculated from electricity generation and electricity price. Third, financial metrics are computed to evaluate investment attractiveness. Fourth, scenario and sensitivity analyses are used to understand uncertainty.

This framework is intentionally simple because the goal is to support investment decision making, not to simulate detailed thermodynamic performance. The model assumes that the ORC system has a rated capacity of 200 kW and that the firm can use or value the generated electricity at a given market price. The model also assumes a project lifetime of 20 years and an 8 percent discount rate.

4. Financial Modeling Methodology

Annual revenue is estimated as the product of annual generation and electricity price:

$$\text{Revenue} = \text{Electricity Generation} \times \text{Electricity Price}$$

Annual generation is calculated as installed capacity multiplied by annual operating hours:

$$\text{Electricity Generation} = \text{Installed Capacity} \times \text{Operating Hours}$$

Return on Investment measures cumulative profit relative to the initial capital cost:

$$\text{ROI} = (\text{Total Profit} / \text{Initial Investment}) \times 100\%$$

Net Present Value discounts future annual cash flows and subtracts the initial investment:

$$\text{NPV} = \sum_{t=1}^n [\text{CF}_t / (1 + r)^t] - I_0$$

where CF_t is annual net cash flow in year t , r is the discount rate, n is the project lifetime, and I_0 is the initial investment.

The payback period measures how long it takes to recover the initial investment through annual net cash flow:

$$\text{Payback Period} = \text{Initial Investment} / \text{Annual Net Cash Flow}$$

These formulas are standard business analytics tools. They allow the project to be explained in financial terms rather than engineering terms.

5. Data and Assumptions

The study uses a transparent illustrative dataset created for investment analysis. The values are not claimed to represent a specific factory. Instead, they represent a reasonable simplified case that can be modified for future real data collection. This is appropriate for a portfolio project because the

objective is to demonstrate the analytical framework and calculation logic.

Variable	Base Value	Interpretation
Installed capacity	200 kW	Rated ORC electricity output
Operating hours	8,000 hours/year	High utilization industrial operation
Electricity price	\$0.15/kWh	Value of generated electricity
Initial investment	\$600,000	\$3,000 per kW installed cost assumption
O&M cost	3% of CAPEX/year	Annual operating and maintenance cost
Project life	20 years	Evaluation horizon
Discount rate	8%	Required return / time value of money

Table 1. Baseline assumptions for the ORC investment analysis.

Three scenarios are constructed to evaluate investment performance under different market assumptions. The conservative case assumes lower electricity prices, fewer operating hours, and higher capital and maintenance costs. The optimistic case assumes stronger electricity prices, high utilization, and lower capital costs.

Scenario	Electricity Price	Operating Hours	Initial Investment	O&M Rate
Conservative	\$0.11/kWh	6,000	\$690,000	4.0%
Base	\$0.15/kWh	8,000	\$600,000	3.0%
Optimistic	\$0.19/kWh	8,500	\$540,000	2.5%

Table 2. Scenario design used for investment evaluation.

6. Results

Under the base case, the 200 kW system operating for 8,000 hours generates 1.6 million kWh per year. At an electricity price of \$0.15 per kWh, the annual gross revenue is \$240,000. After annual O&M; cost of \$18,000, annual net cash flow equals \$222,000. This creates a positive NPV over the 20-year project life.

Scenario	Annual Revenue	Annual Net CF	NPV	ROI	Payback
Conservative	\$132,000	\$104,400	\$335,015	202.6%	6.6 yrs
Base	\$240,000	\$222,000	\$1,579,629	640.0%	2.7 yrs
Optimistic	\$323,000	\$309,500	\$2,498,717	1046.3%	1.7 yrs

Table 3. Investment results by scenario.

The conservative case remains weaker because the lower electricity price and reduced operating hours limit annual revenue. The base case produces a strong positive NPV, while the optimistic case performs best because higher electricity price and utilization increase cash flow. These results show why ORC adoption should be evaluated using both technical feasibility and economic context.

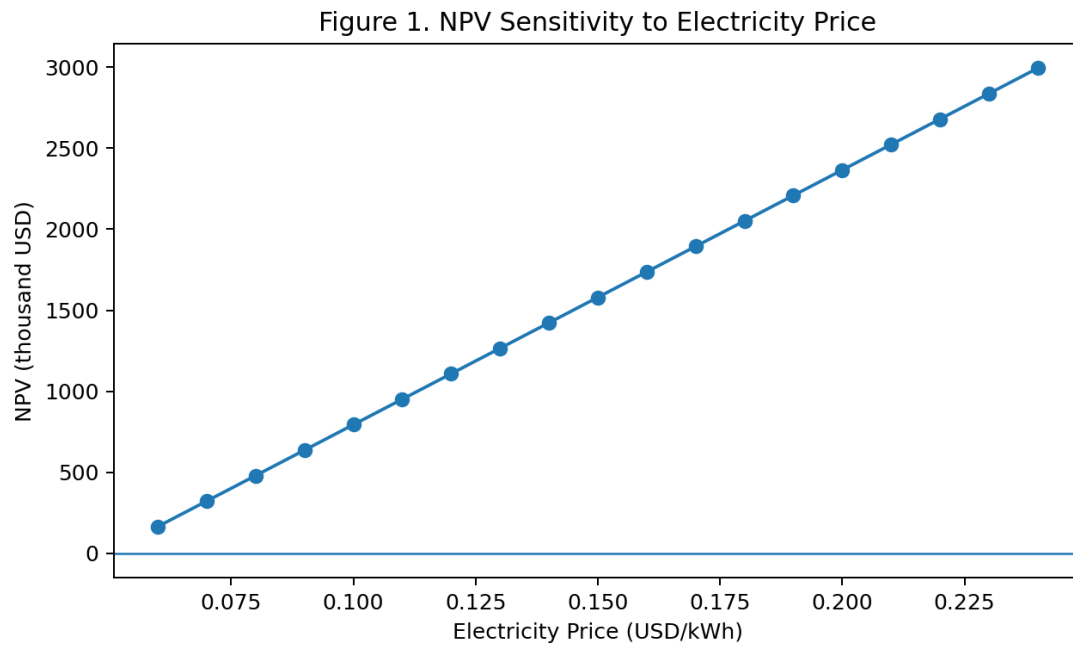


Figure 1. NPV increases as electricity price rises. The break-even point occurs when NPV crosses zero.

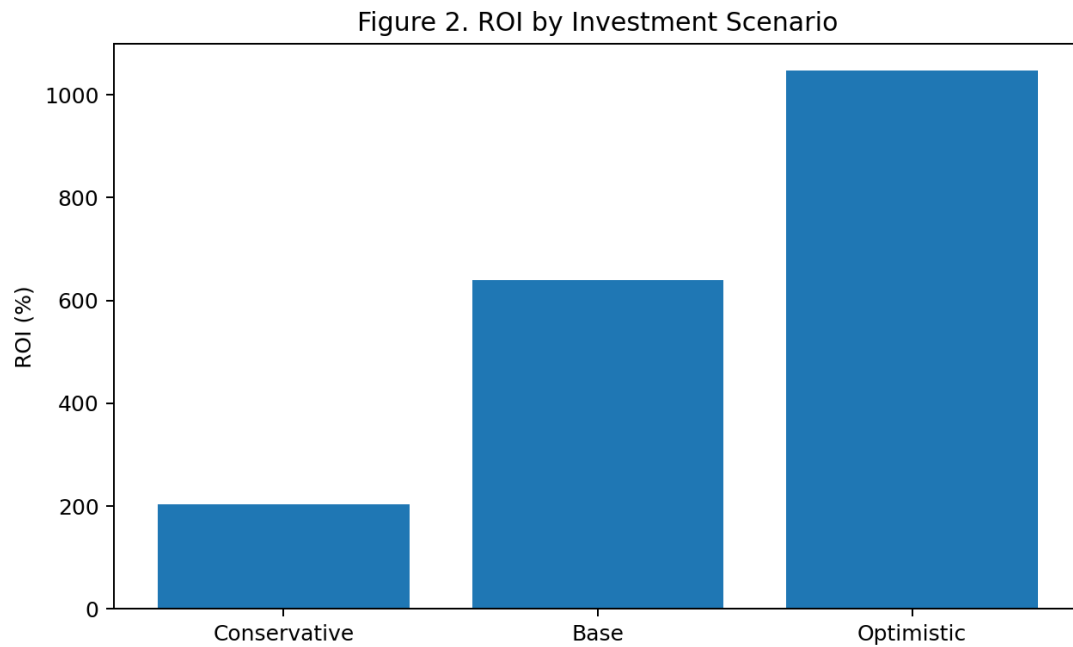


Figure 2. ROI improves from conservative to optimistic scenarios.

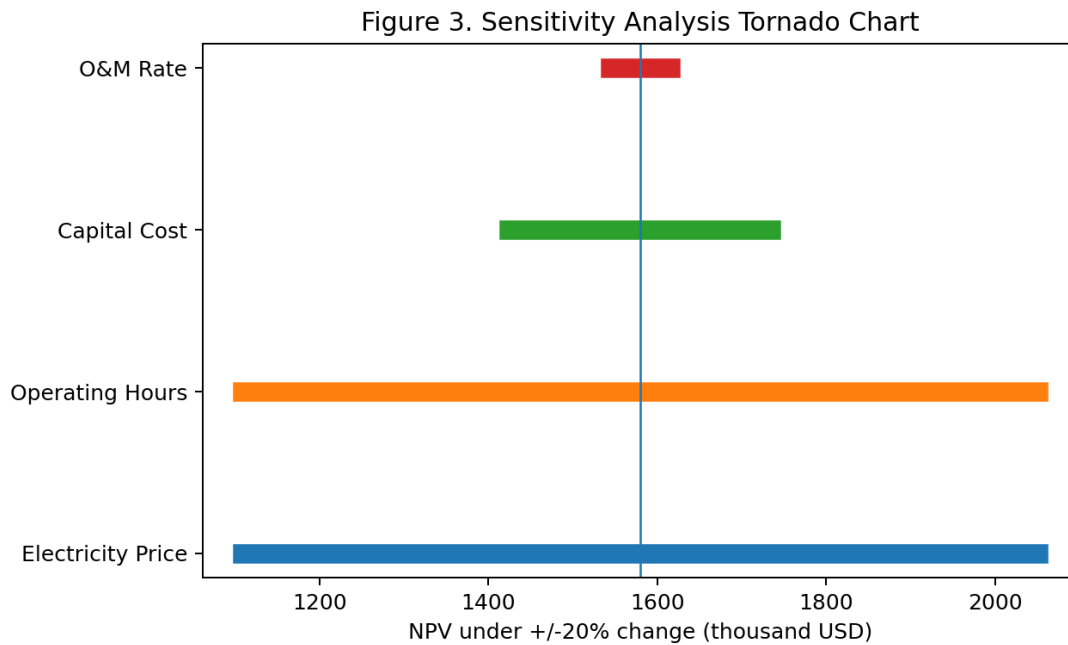


Figure 3. Sensitivity analysis shows which variables have the largest impact on NPV.

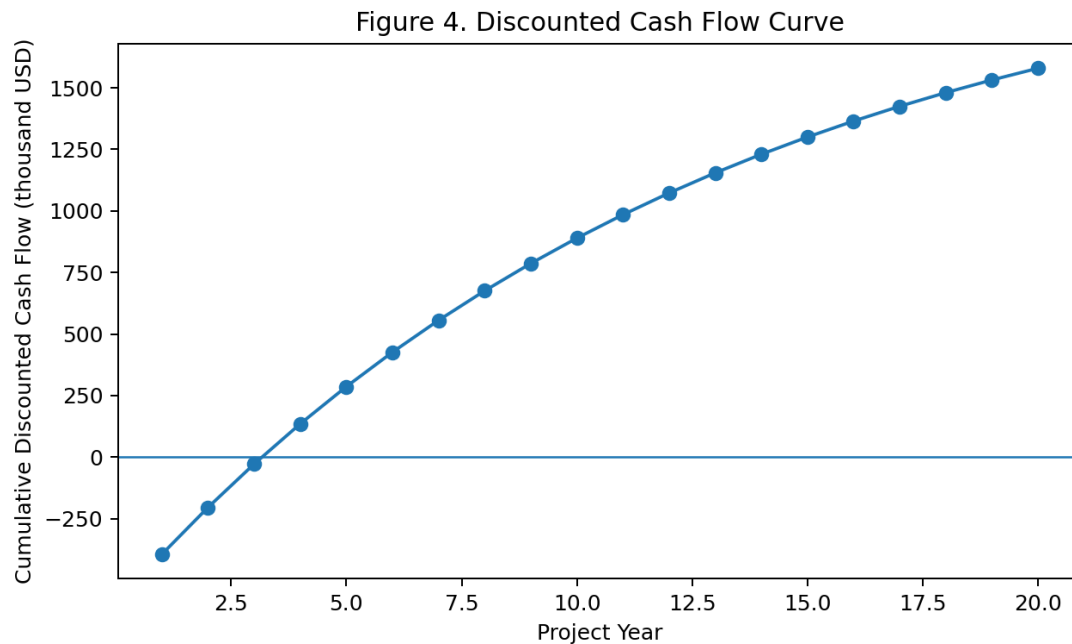


Figure 4. Cumulative discounted cash flow becomes positive after the project recovers the initial investment.

7. Sensitivity Analysis

Sensitivity analysis changes one assumption at a time by plus or minus 20 percent. This allows the study to identify the variables that matter most for profitability. Electricity price has the largest impact because every additional cent per kWh directly increases annual revenue. Operating hours also matter because higher utilization spreads the initial investment over more generated electricity. Capital cost matters because it determines the initial investment that must be recovered.

The sensitivity results support a clear managerial interpretation: before adopting ORC, a firm should first evaluate local electricity prices and expected operating hours. A factory with stable year-round heat availability is more likely to benefit than a facility with intermittent operations.

8. Discussion and Managerial Implications

The results suggest that ORC investment should be treated as a structured capital budgeting problem. Managers should not approve a project simply because it is energy efficient. They should compare expected cash flows, discount future benefits, and test whether the investment remains attractive under conservative assumptions.

This study also demonstrates how business analytics can translate engineering opportunities into financial decision support. The same framework can be applied to other sustainability investments, such as solar installations, battery storage, process efficiency upgrades, and industrial automation projects. The value of analytics is not only producing a single estimate, but also showing how conclusions change when assumptions change.

9. Limitations and Future Research

This paper uses a simplified illustrative dataset rather than operational data from a specific industrial plant. It assumes stable electricity prices, constant operating hours, and fixed annual O&M rates. The model also does not include tax incentives, carbon credits, equipment degradation, financing structure, or policy subsidies. These limitations are acceptable for a first-stage business analytics portfolio project but should be addressed in future research.

Future work could extend the model using Monte Carlo simulation, probability distributions for electricity prices, machine learning forecasts for energy markets, and multi-site portfolio optimization. These extensions would make the analysis more statistical and better suited for larger-scale investment planning.

10. Conclusion

This study developed a business analytics framework for evaluating industrial ORC investment feasibility. Using revenue estimation, ROI, NPV, payback period, scenario analysis, and sensitivity analysis, the paper shows that electricity price, annual operating hours, and capital cost are the main drivers of financial attractiveness. The project demonstrates the ability to frame a technical industrial opportunity as a quantitative investment decision. For PhD application purposes, the contribution is in financial modeling, data-driven decision support, and interpretation of uncertainty, rather than detailed thermal engineering design.

11. Step-by-Step Numerical Interpretation

This section explains the base case using plain business logic. A 200 kW system operating for 8,000 hours per year generates 1,600,000 kWh annually. When each kWh is valued at \$0.15, the gross annual revenue is \$240,000. This is not yet profit because the project must pay operating and maintenance expenses. With annual O&M equal to 3 percent of the \$600,000 initial investment, annual O&M equals \$18,000. Therefore, annual net cash flow is \$222,000.

The ROI calculation asks whether the total profit across the project life is large relative to the amount invested at the beginning. In the base case, cumulative net cash flow over 20 years is \$4,440,000 before subtracting the original \$600,000 investment. This creates total profit of \$3,840,000, which produces a high ROI. This value should not be interpreted as a guarantee. It is a result of the assumptions used in the illustrative dataset.

The NPV calculation is more conservative because it recognizes that money received in future years is worth less than money received today. Each annual cash flow is discounted by 8 percent before being added to the project value. If the discounted value of future cash flows is greater than the initial investment, the NPV is positive. A positive NPV means the project is expected to create value after considering the required return.

The payback period provides an easier but less complete view. It divides the initial investment by annual net cash flow. In the base case, this means the project recovers its initial cost in less than three years. However, payback period does not consider cash flows after the payback year and does not discount future cash flows. Therefore, it should be used together with NPV rather than alone.

12. Practical Decision Rule for Managers

Based on this analysis, a manager can use a simple decision rule. First, estimate expected annual operating hours. If the ORC system cannot operate for a large portion of the year, the investment becomes less attractive. Second, estimate the electricity price or avoided electricity cost. If the value of electricity is low, revenue will not be sufficient to recover the capital cost. Third, calculate NPV under conservative assumptions. If the conservative case is negative but the base case is strongly positive, the project may still be worth additional due diligence. If all cases are negative, the investment should likely be rejected or redesigned.

This decision rule is useful because it prevents managers from relying on a single optimistic forecast. It also supports transparent communication between technical teams and financial decision makers. Engineering teams can provide expected capacity and operating conditions, while analytics teams translate those inputs into financial metrics.

13. Relevance to Data Science and Business Analytics

Although this paper is about an industrial energy technology, the analytical contribution is broader. The project demonstrates how to structure an ambiguous technical investment problem into a measurable decision model. This type of work is relevant to data science, statistics, and information systems because organizations increasingly need analysts who can combine data, assumptions, uncertainty, and business interpretation.

The current model is deterministic, but it naturally leads to statistical extensions. For example, electricity price can be modeled as a random variable, annual operating hours can be modeled using historical production data, and capital cost uncertainty can be represented using probability distributions. These extensions would allow Monte Carlo simulation and confidence intervals for investment outcomes. In this way, the project can evolve from a simple business analytics study into a more advanced statistical decision-support system.

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